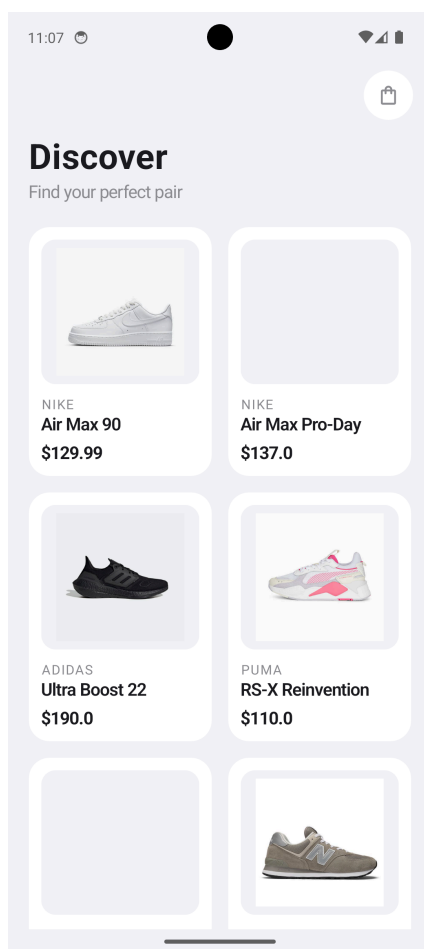


ShoeShop

Kotlin Jetpack Compose Mobile Shopping App



Course: Mobile Application Development

Framework: Kotlin Jetpack Compose (Material3)

Platform: Android (API 26+)

Date: April 30, 2026

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1 Project Overview

ShoeShop is a mobile shopping application built with **Kotlin Jetpack Compose** that demonstrates modern Android UI development. The app allows users to:

- Browse a catalog of shoes displayed in a responsive grid layout
- View detailed product information including description, price, and available sizes
- Add items to a shopping bag with quantity management
- View shopping bag total with itemized pricing

1.1 Key Design Principles

- **Clean UI** — Minimal typography, generous spacing, subtle shadows
- **Material3 Design System** — Using the latest Material Design components
- **Smooth Animations** — Page transitions and press-state feedback
- **Edge-to-Edge** — Full immersive display support

2 How to Configure and Run

2.1 Prerequisites

- Android Studio Hedgehog (2023.1.1) or later
- JDK 17
- Android SDK 34
- Android Emulator or physical device (API 26+)

2.2 Steps to Run

1. Open Android Studio
2. Select **File** > **Open** and choose the **ShoeShop** folder
3. Wait for Gradle sync to complete (first time may take a few minutes)
4. Create a virtual device: **Tools** > **Device Manager** > **Create Virtual Device**
 - Select **Pixel 6** (or any phone)
 - Choose system image **API 34**
 - Click **Finish**
5. Select the emulator from the device dropdown
6. Click the **Run** button (green play icon) or press **Shift+F10**

3 Project Structure

```

ShoeShop/
|-- app/
|   |-- build.gradle.kts          (App-level build config)
|   +-- src/main/
|       |-- AndroidManifest.xml  (App manifest)
|       |-- java/com/example/shoeshop/
|           |-- MainActivity.kt   (Entry point & navigation)
|           |-- model/
|               |-- Shoe.kt       (Shoe data class)
|               +-- CartItem.kt   (Cart item data class)
|               |-- data/
|                   +-- ShoeData.kt (Static shoe catalog)
|               +-- ui/
|                   |-- theme/
|                       |-- Color.kt (Color definitions)
|                       +-- Theme.kt (Material3 theme)
|                   |-- components/
|                       +-- ShoeCard.kt (Product card widget)
|                   +-- screens/
|                       |-- HomeScreen.kt (Product grid)
|                       |-- DetailScreen.kt (Product detail)
|                       +-- CartScreen.kt (Shopping bag)
|               +-- res/values/
|                   |-- strings.xml
|                   +-- themes.xml
|-- build.gradle.kts              (Root build config)
|-- settings.gradle.kts          (Project settings)
+-- gradle/wrapper/
    +-- gradle-wrapper.properties (Gradle version)

```

4 Technologies Used

Technology	Purpose
Kotlin	Programming language
Jetpack Compose	Declarative UI framework
Material3	Design system & components
Coil	Asynchronous image loading
Compose Animation	Page transitions & interactions
Edge-to-Edge API	Immersive full-screen display

5 Features

5.1 Home Screen

Displays shoes in a 2-column responsive grid with:

- Product images loaded asynchronously from network
- Brand name (uppercase with letter spacing)
- Product name and price
- Press animation feedback (scale to 96%)
- Shopping bag icon with item count badge

5.2 Detail Screen

Shows full product information:

- Large product image with white background
- Brand, name, and description
- Size selector (7–11)
- Floating back and favorite buttons
- Bottom bar with price and “Add to Bag” button

5.3 Cart Screen

Shopping bag management:

- Item cards with thumbnail, name, brand, price
- Quantity controls (increment/decrement)
- Per-item total calculation
- Subtotal, shipping (free), and grand total
- Checkout button

6 Screenshots

6.1 Home Screen

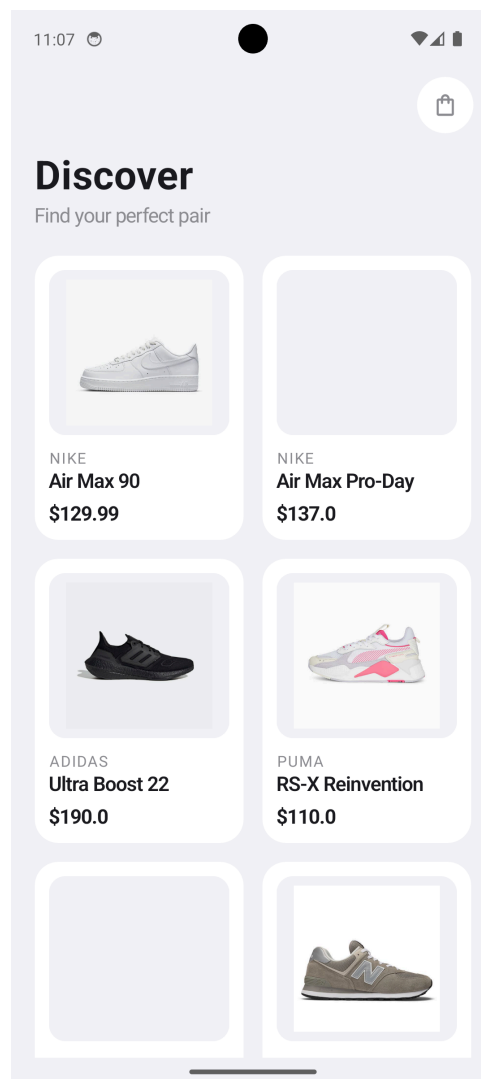


Figure 1: Product catalog with 2-column grid layout

6.2 Detail Screen

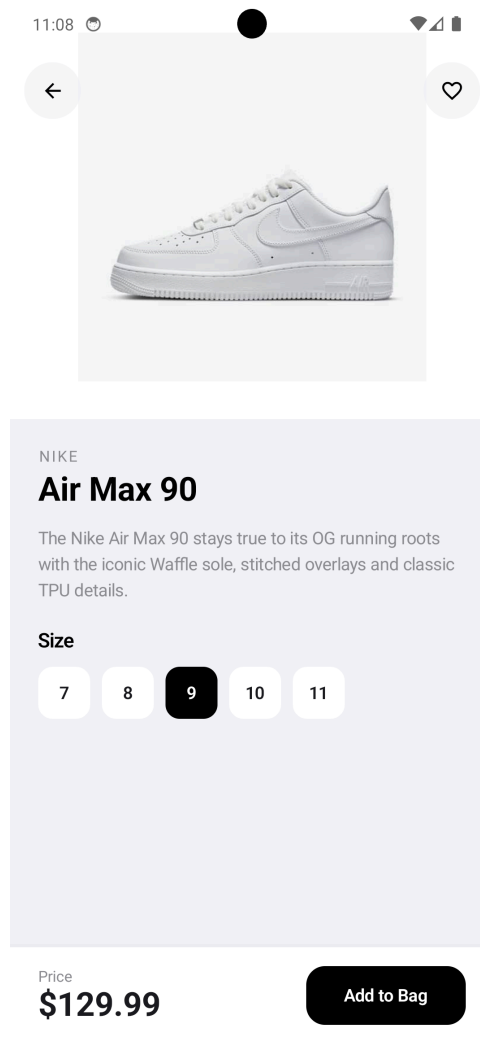


Figure 2: Product detail with size selector and Add to Bag

6.3 Shopping Cart — Single Item

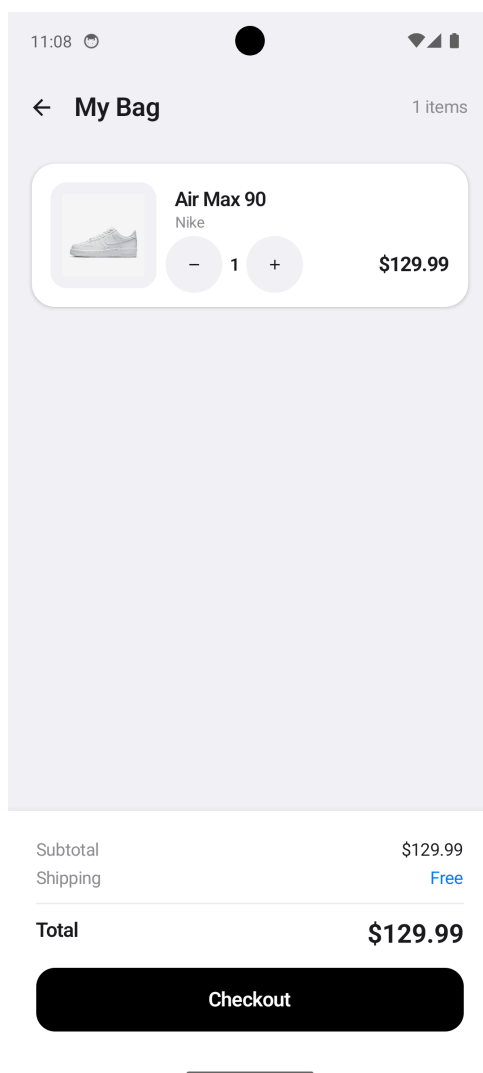


Figure 3: Shopping bag with one item and total calculation

6.4 Shopping Cart — Multiple Items

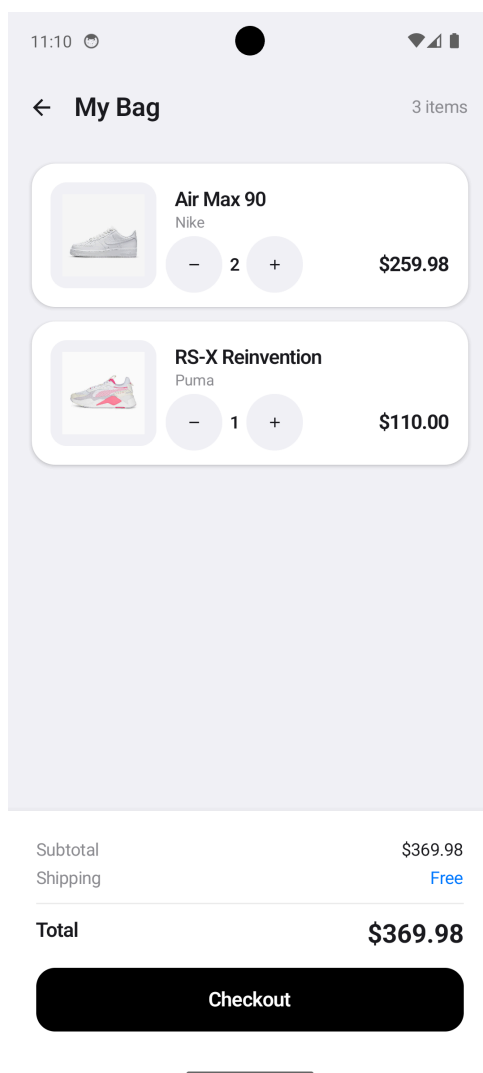


Figure 4: Shopping bag with multiple items and quantity adjustment